

QuantAgent - App Summary

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One-page summary generated from repository evidence only

What It Is

QuantAgent is an AI-assisted quantitative trading app that combines a Python multi-agent backend with dashboard UIs. In this repo it supports research, signal generation, risk review, paper execution, backtesting, and monitoring for US and A-share symbols.

Who It's For

Primary persona appears to be a quant trader or strategy researcher running local paper-trading workflows. Formal user-persona documentation is **Not found in repo**.

What It Does

- Orchestrates Researcher, Strategist, Risk Manager, and Executor agents through a coordinator loop.
- Pulls market history, stock info, and news through a yfinance-backed provider, with Google News RSS as a supplement.
- Computes technical indicators and simple title-based sentiment for symbol analysis.
- Generates ranked signals with confidence, entry, target, stop-loss, and position-size metadata.
- Applies paper-trading controls such as signal approval/rejection, manual orders, position close, reset, and auto-trading settings.
- Exposes REST and WebSocket feeds for portfolio, markets, signals, trades, risk, alerts, logs, charts, news, and watchlists.
- Runs CLI backtests and includes both a React dashboard and a legacy Streamlit dashboard.

How It Works

React dashboard (frontend/)
-> GET /api/* and ws://localhost:8000/ws
-> FastAPI app (quant_agent/api/main.py)
-> DashboardRuntimeService per market + strategy
-> Researcher / Strategist / RiskManager / Executor
-> ToolRegistry + AgentMemory + RuntimeStateStore
-> YFinanceMarketDataProvider for prices/info/news
-> SQLite runtime state at data/runtime_state.db

CLI paths:

quant_agent.cli:start -> CoordinatorAgent loop
quant_agent.cli:run-backtest -> BacktestEngine -> provider history

Notes: the runtime service keeps a synthetic fallback snapshot via DashboardDemoService for local development. Broker routing / production deployment topology is **Not found in repo**.

How To Run

1. pip install -r requirements.txt
2. cp .env.example .env and add keys as needed.
3. Start the API: python -m quant_agent serve-api --reload
4. Start the React UI: cd frontend && npm install && npm run dev
5. Open http://localhost:3000.
6. Optional legacy UI: python -m quant_agent dashboard

Evidence Base

Summary derived from repo files including README.md, SETUP.md, quant_agent/cli.py, quant_agent/api/main.py, quant_agent/api/runtime_service.py, quant_agent/backtest/engine.py, quant_agent/core/tools.py, quant_agent/data/providers/yfinance_provider.py, and frontend/src/hooks/useApp.tsx.